

21.1: Equations for Projectiles

Uses SUVAT, but horizontal & vertical systems are treated separately in vector form.

Hence:

$\vec{v_0} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} v_0 \cos\theta \\ v_0 \sin\theta \end{pmatrix}$. In general: $\vec{a} = \begin{pmatrix} 0 \\ -g \end{pmatrix}$

$\therefore$

$\vec{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} + \begin{pmatrix} 0 \\ -g \end{pmatrix} t$ & $\vec{x} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} t + \frac{1}{2}\begin{pmatrix} 0 \\ -g \end{pmatrix} t^2$

At the maximum height, $v_y = 0$, and the particle lands when $x_y = 0$.
Time of flight is twice that from the origin to the peak, such that the particle lands at $y = 0$
For $a_x = 0$, Range $= u_x T$.

21.4: The Path of a Projectile:

General projectile equations use the parameter of time. This can hence be removed (using the x equation) and then be resubstituted (into the y equation). This shows the direct relationship between the horizontal & vertical positions

21.5: General Equations:

$\vec{a} = \begin{pmatrix} 0 \\ -g \end{pmatrix}$ $\vec{u} = \begin{pmatrix} u\cos\theta \\ u\sin\theta \end{pmatrix}$

$\Rightarrow x = ut\cos\theta \Rightarrow t = \frac{x}{u\cos\theta}$

$\Rightarrow y = ut\sin\theta - \frac{1}{2}gt^2 = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\Rightarrow y = x\tan\theta - \frac{gx^2}{2u^2}(\tan^2\theta + 1)$

This is the general equation for $\vec{x}$.

Max Height & Time

$v_y = u\sin\theta - gt = 0$
$\Rightarrow \frac{u\sin\theta}{g} = t$

$\Rightarrow y = \frac{u^2\sin^2\theta}{g} - \frac{u^2\sin^2\theta}{2g}$
$= \frac{u^2\sin^2\theta}{2g}$ MH

$0 = ut\sin\theta - \frac{1}{2}gt^2$
$= t(u\sin\theta - \frac{1}{2}gt)$
$\Rightarrow t = 0$,
$t = \frac{2u\sin\theta}{g}$ ToF

$R = \frac{2u^2\sin\theta\cos\theta}{g}$
$= \frac{u^2\sin 2\theta}{g}$